

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Simulate smoothing and sharpening operation on images using spatial filters.	
Experiment No: 9	Date:	Enrollment No: 92510133028

write a python program to simulate smoothing and sharpening operation on image using spatial filter

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files

# Step 1: Upload the image
print("Please upload an image file (e.g., Untitled.jpg)")
uploaded = files.upload()

if not uploaded:
    print("No file uploaded.")
else:
    filename = list(uploaded.keys())[0]
    # Load the image
    image = cv2.imread(filename)

    if image is None:
        print("Error: Could not decode the image.")
    else:
        # Define the Gaussian kernel for smoothing
        kernel_size = (5, 5)
        sigma = 1.5
        gaussian_kernel = cv2.getGaussianKernel(kernel_size[0], sigma)
        gaussian_kernel = np.outer(gaussian_kernel, gaussian_kernel)

        # Apply Gaussian smoothing
        smoothed_image = cv2.filter2D(image, -1, gaussian_kernel)

        # Define a sharpening kernel
        sharpening_kernel = np.array([[ -1, -1, -1],
                                       [ -1,  9, -1],
                                       [ -1, -1, -1]])

        # Apply sharpening
        sharpened_image = cv2.filter2D(image, -1, sharpening_kernel)

        # Display the original image, smoothed, and sharpened images
        print("Original Image:")
        cv2_imshow(image)
```

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```
print("Smoothed Image (Gaussian):")
cv2_imshow(smoothed_image)
print("Sharpened Image:")
cv2_imshow(sharpened_image)
```



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write a python program to simulate smoothening using Averaging linear filters

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files

# Step 1: Upload the image
print("Please upload an image file for Averaging Filter:")
uploaded = files.upload()

if not uploaded:
    print("No file uploaded.")
else:
    filename = list(uploaded.keys())[0]
    # Load the image
    image = cv2.imread(filename)
```

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```

if image is None:
    print("Error: Could not decode the image.")
else:
    # Define the size of the Averaging filter kernel
    kernel_size = (5, 5)

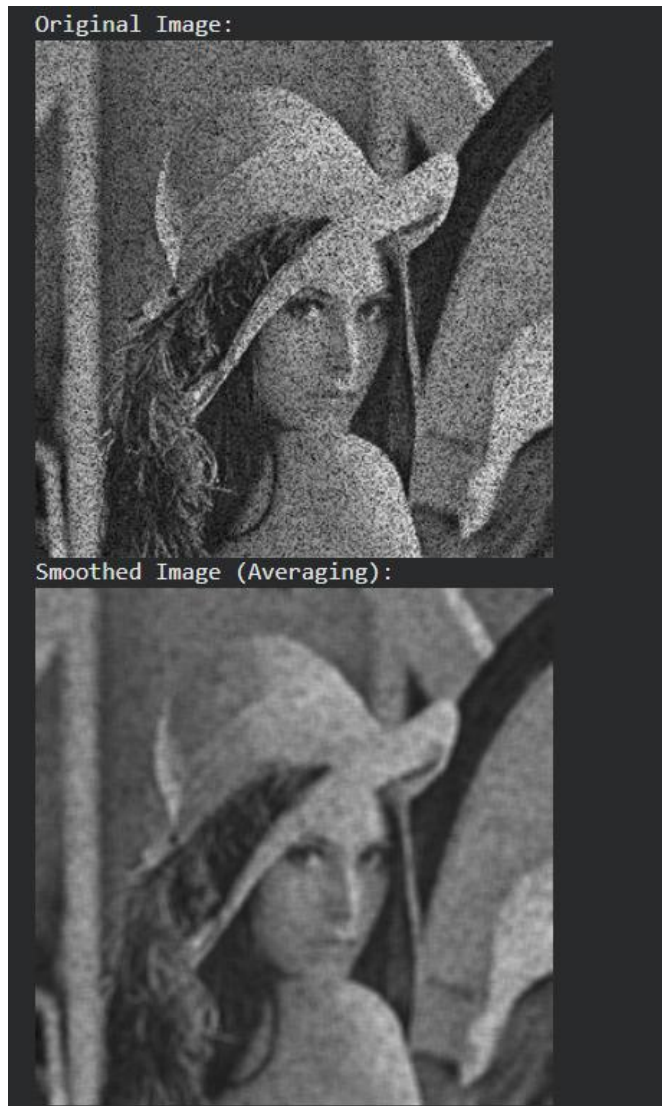
    # Create the Averaging filter kernel
    kernel = np.ones(kernel_size, dtype=np.float32) /
(kernel_size[0] * kernel_size[1])

    # Apply the Averaging filter for smoothing
    smoothed_image = cv2.filter2D(image, -1, kernel)

    # Display the original and smoothed images
    print("Original Image:")
    cv2_imshow(image)
    print("Smoothed Image (Averaging):")
    cv2_imshow(smoothed_image)

```

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write a python program to simulate smoothening using median filters

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files

# Step 1: Upload the image
print("Please upload an image file for Median Filter:")
```

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Experiment No: 9	Date:	Enrollment No: 92510133028

```

uploaded = files.upload()

if not uploaded:
    print("No file uploaded.")
else:
    filename = list(uploaded.keys())[0]
    # Load the image
    image = cv2.imread(filename)

    if image is None:
        print("Error: Could not decode the image.")
    else:
        # Define the size of the median filter kernel (should be an odd
number)
        kernel_size = 5

        # Apply the Median filter for smoothing
        smoothed_image = cv2.medianBlur(image, kernel_size)

        # Display the original and smoothed images
        print("Original Image:")
        cv2_imshow(image)
        print("Smoothed Image (Median Filter):")
        cv2_imshow(smoothed_image)

```


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write a python program to simulate sharpening using spatial high pass filters

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files

# Step 1: Upload the image
```

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```

print("Please upload an image file for Laplacian Sharpening:")
uploaded = files.upload()

if not uploaded:
    print("No file uploaded.")
else:
    filename = list(uploaded.keys())[0]
    # Load the image
    image = cv2.imread(filename)

    if image is None:
        print("Error: Could not decode the image.")
    else:
        # Apply Gaussian smoothing to reduce noise (optional but recommended)
        blurred_image = cv2.GaussianBlur(image, (5, 5), 0)

        # Create a Laplacian kernel for sharpening
        laplacian_kernel = np.array([[0, -1, 0],
                                     [-1, 5, -1],
                                     [0, -1, 0]], dtype=np.float32)

        # Apply the Laplacian filter for sharpening
        sharpened_image = cv2.filter2D(blurred_image, -1, laplacian_kernel)

        # Display the results
        print("Original Image:")
        cv2_imshow(image)
        print("Blurred Image (Pre-processing):")
        cv2_imshow(blurred_image)
        print("Sharpened Image (Laplacian):")
        cv2_imshow(sharpened_image)

```


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Original Image:



Blurred Image (Pre-processing):



Sharpened Image (Laplacian):



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Conclusion

From this experiment, I learned how to perform smoothing and sharpening of images using spatial filters. I understood that smoothing filters help reduce noise and unwanted details, while sharpening filters enhance edges and fine details. This experiment helped me understand how spatial filtering can be used to improve the quality and appearance of images.

https://github.com/keshvi1234/DSIP_Exp